

PRACTICAL 7 (A) – DHCP

1. Devices Required

You need **8 devices**:

Network Devices

1. **Router0**
2. **Switch0**

End Devices

3. **Server0**
4. **PC0 , PC1, PC2 , PC3 , PC4**

Arrange the Devices

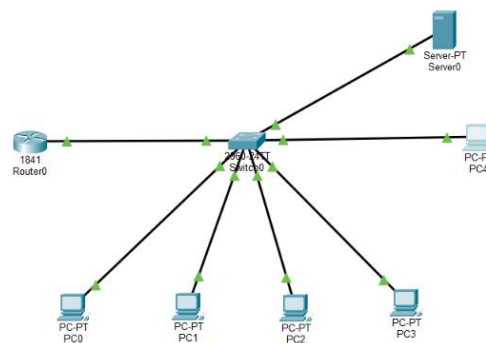
Connect the Copper Wires

Select:

Connections ⚡ → **Copper Straight-Through**

Make these **7 connections**:

From Device	Port	To Device	Port
Router0	FastEthernet 0/0	Switch0	FastEthernet 0/1
Server0	FastEthernet 0	Switch0	FastEthernet 0/2
PC0	FastEthernet 0	Switch0	FastEthernet 0/3
PC1	FastEthernet 0	Switch0	FastEthernet 0/4
PC2	FastEthernet 0	Switch0	FastEthernet 0/5
PC3	FastEthernet 0	Switch0	FastEthernet 0/6
PC4	FastEthernet 0	Switch0	FastEthernet 0/7



- **Configure Router0**

Click:

Router0 → Config → FastEthernet 0/0

Enter:

IP Address: 10.0.0.1

Subnet Mask: 255.0.0.0

Set:

Port Status → On

- **Configure Server0**

Go to:

Server0 → Desktop → IP Configuration

Enter:

IP Address: 10.0.0.2

Subnet Mask: 255.0.0.0

Default Gateway: 10.0.0.1

DNS Server: 10.0.0.2

- **Configure DHCP on Server0**

Go to:

Server0 → Services → DHCP

Turn:

DHCP → ON

Enter the following:

Setting	Value
Pool Name	DHCP
Default Gateway	10.0.0.1
DNS Server	10.0.0.2
Start IP Address	10.0.0.3
Subnet Mask	255.0.0.0
Maximum Users	512

Then click **Add**.

- **Configure PC0 for DHCP**

Click:

PC0 → Desktop → IP Configuration

Select:

DHCP

Wait a few seconds.

PC0 should automatically receive an IP address.

- **Configure PC1 for DHCP**

Go to:

PC1 → Desktop → IP Configuration

Select:

DHCP

PC1 will automatically receive an IP address.

- **Configure PC2 for DHCP**

Go to:

PC2 → Desktop → IP Configuration

Select:

DHCP

PC2 will automatically receive an IP address.

- **Configure PC3 for DHCP**

Go to:

PC3 → Desktop → IP Configuration

Select:

DHCP

PC3 will automatically receive an IP address.

- **Configure PC4 for DHCP**

Go to:

PC4 → Desktop → IP Configuration

Select:

DHCP

PC4 will automatically receive an IP address.

PRACTICAL 7 (B) - ROUTER AS DHCP SERVER

1. Devices Required

You need:

1 × Cisco 2811 Router → Router0

2 × Cisco 2950-24 Switches → Switch0, Switch1

4 × PCs → PC0, PC1, PC2, PC3

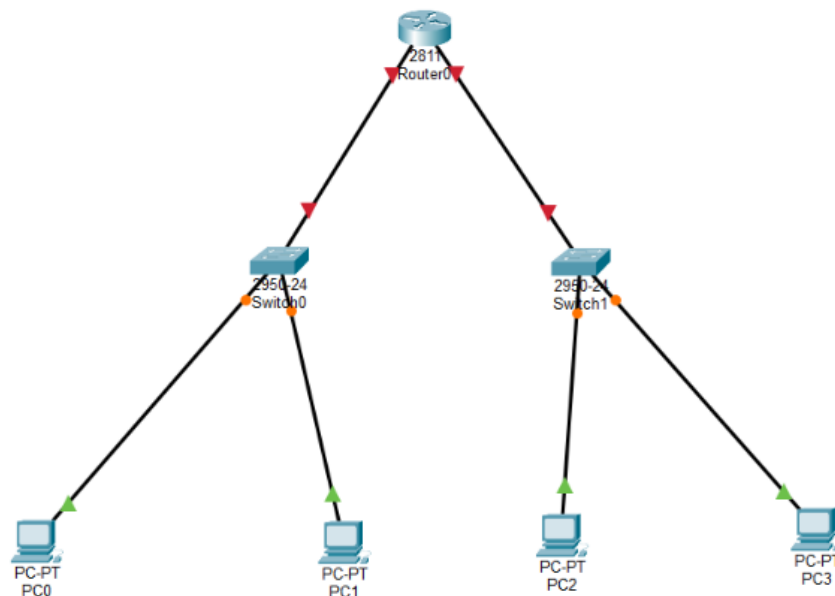
Connect the Copper Wires

Select:

Connections ⚡ → **Copper Straight-Through**

Make exactly these connections:

From	Port	To	Port
Router0	Fa0/0	Switch0	Fa0/24
Router0	Fa0/1	Switch1	Fa0/24
Switch0	Fa0/1	PC0	FastEthernet0
Switch0	Fa0/2	PC1	FastEthernet0
Switch1	Fa0/1	PC2	FastEthernet0
Switch1	Fa0/2	PC3	FastEthernet0



Wait until the links become green.

- **Configure Router0 — Fa0/0**

Click:

Router0 → Config → FastEthernet 0/0

Set:

IP Address: 192.168.0.1

Subnet Mask: 255.255.255.0

Turn:

Port Status → On

Configure Router0 — Fa0/1

Still on Router0, select:

Config → FastEthernet 0/1

Set:

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Turn:

Port Status → On

- **Configure DHCP Pool for the Left Network**

Click:

Router0 → CLI

Enter:

enable

configure terminal

Then:

ip dhcp pool LAN0

Your prompt should change to:

Router(dhcp-config)#

Now enter:

network 192.168.0.0 255.255.255.0

default-router 192.168.0.1

exit

- **Configure DHCP Pool for the Right Network**

Now enter:

```
ip dhcp pool LAN1
```

Then:

```
network 192.168.1.0 255.255.255.0
```

```
default-router 192.168.1.1
```

```
exit
```

Finally:

```
end
```

```
write memory
```

- **Configure PC0 for DHCP**

Click:

PC0 → Desktop → IP Configuration

Select:

DHCP

Do **not** manually enter an IP address.

- **Configure PC1 for DHCP**

Go to:

PC1 → Desktop → IP Configuration → DHCP

PC1 should receive another 192.168.0.x address.

- **Configure PC2 for DHCP**

Go to:

PC2 → Desktop → IP Configuration → DHCP

PC2 should receive a 192.168.1.x address.

- **Configure PC3 for DHCP**

Go to:

PC3 → Desktop → IP Configuration → DHCP

PC3 should receive another 192.168.1.x address.